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RETRACTING TISSUE

Abstract

Surgical retractors and methods thereof for retracting body tissue in a therapeutic procedure. The surgical retractor may include retractor blades have openings or channels configured to receive bone pins while maintaining accessibility and visibility of the operative window.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. application Ser. No. 18/402,887, filed on Jan. 3, 2024, which is a continuation of U.S. application Ser. No. 17/537,577 filed on Nov. 30, 2021 (published as U.S. Pat. Pub. No. 2022-0079703), which is a continuation of U.S. application Ser. No. 16/364,268 filed on Mar. 26, 2019, now U.S. Pat. No. 11,234,788, which is a divisional of U.S. application Ser. No. 15/019,001 filed on Feb. 9, 2016, now U.S. Pat. No. 10,278,786, which is a continuation-in-part of U.S. application Ser. No. 14/664,089, filed Mar. 20, 2015, now U.S. Pat. No. 10,039,539, which is a continuation-in-part application of 14/183,048, filed Feb. 18, 2014, now U.S. Pat. No. 10,285,680, all of which are hereby incorporated by reference in their entireties for all purposes.

FIELD OF THE INVENTION

[0002] The invention relates to a system and method for retracting body tissue during surgery, and more particularly to multi-axis retractors affixed to a base.

BACKGROUND OF THE INVENTION

[0003] Retractor systems may be used in a variety of different surgical procedures to provide an opening through which a doctor may access the surgical site. In spinal surgeries, for example, a retractor system may be used to provide the surgeon with access to the patient's spine. The opening created by the retractor system may, for example, enable the doctor to insert surgical instruments into the body or enable visualization of the surgical site using visible light or X-rays.

[0004] A retractor system may include a plurality of blades coupled to a retractor frame. In use, the blades may be inserted into an incision and then retracted to displace tissue surrounding the incision along a path to the surgical site. To minimize trauma to the tissue, this tissue displacement should generally be refined and controlled.

SUMMARY OF THE INVENTION

[0005] In accordance with an embodiment of the disclosure, a surgical retractor for retracting body tissue in a therapeutic procedure comprises at least one first arm defining proximal and distal ends and a first arm longitudinal axis extending therebetween, the distal end configured to support a retractor blade, the proximal end forming one of an extension or a block having an aperture; at least one second arm defining proximal and distal ends and a second arm longitudinal axis extending therebetween, the distal end configured to support a retractor blade, the proximal end forming one of an extension or a block having an aperture; a core: (a) configured to form a slideable connection each with the at least one first arm and the at least one second arm, (b) the slideable connection with the at least one first arm formed as a block having an aperture when the proximal end of the at least one first arm forms an extension, the slideable connection formed as an extension when the proximal end of the at least one first arm forms a block having an aperture, (c) the slideable connection with the at least one second arm formed as a block having an aperture when the proximal end of the at least one second arm forms an extension, the slideable connection formed as an extension when the proximal end of the at least one second arm forms a block having an

aperture, (d) each first arm translatable using the slideable connection independently of translation of at least one second arm, (e) each second arm translatable using the slideable connection independently of translation of at least one first arm, and (f) the distal end of at least one first arm and the distal end of at least one second arm mutually separable using the slideable connection to thereby separate retractor blades supported upon the at least one first arm and the at least one second arm.

[0006] In variations thereof, each first arm is translatable using the slideable connection along an axis transverse to the first longitudinal axis, and each second arm is translatable using the slideable connection along the second longitudinal axis; there are two first arms and one second arm; the two first arms and one second arm formable into a cannula; one or more retractor blades have a tapered end configured to retract tissue that is bone; at least one retractor blade is pivotally mounted to at least one of the at least one first arm and the at least one second arm; pivoting the at least one retractor blade retracts tissue, and separating at least one first arm from at least one second arm retracts tissue; and/or at least one retractor blade is pivoted by rotating a threaded shaft.

[0007] In further variations thereof, the retractor further includes at least one blade holder pivotally mounted to at least one of the at least one first arm and the at least one second arm; a blade is connected to each of the at least one blade holder using a dovetail connection; and/or a light source is affixable to at least one retractor blade.

[0008] In yet further embodiments thereof, the therapeutic procedure is conducted in a surgical facility having a supporting surface including an extension arm, the retractor further including a mount moveably connected to the core and configured to releasably connect to the extension arm; and/or the mount is moveably connected to the core by a pivot.

[0009] In other variations thereof, the slideable connection between at least one of the first and second arms and the core includes a rack and a mating ratcheting pawl, the rack and pawl configured to control relative movement of the at least one first and second arm and the core; the retractor further includes a pinion rotatably secured to one of the at least one first and second arms and the core, the rack associated with the other of the at least one first and second arm, the pinion rotatable to change a relative position of the at least one first and second arm and the core; and/or the retractor further includes at least one of a manually engageable key and a tool engagement connected to the pinion, the key or the tool engagement rotatable to rotate the pinion.

[0010] In other embodiments thereof, the retractor further includes a blade holder including a block affixed to at least one of the at least one first and second arms and defining an aperture and an extending flange; a blade retainer having an extension sized and dimensioned to be insertable into the aperture; a threadable set screw connectable between the blade retainer and the extending flange to retain the blade retainer in connection with the block; and/or the retractor further includes a blade holder including a block affixed to at least one of the at least one first and second arms and defining an aperture and an extending flange; a retractor blade having an extension sized and dimensioned to be insertable into the aperture; a threadable set screw connectable between the retractor blade and the extending flange to retain the retractor blade in connection with the block.

[0011] In another embodiment of the disclosure, a surgical retractor for retracting body tissue in a therapeutic procedure comprises at least one first arm defining proximal and distal ends and a first arm longitudinal axis extending therebetween, the proximal end defining a first block having a first aperture, the first aperture defining a first aperture longitudinal axis that is transverse to the first arm longitudinal axis, the distal end configured to support a retractor blade; a transverse extension extending through and slideably supporting at least one first block of the at least one first arm; at least one second arm defining proximal and distal ends and a second arm longitudinal axis extending therebetween, the proximal end defining a second extension, the distal end configured to support a retractor blade; and a second block connected to the transverse extension and having a second aperture sized and dimensioned to slideably support at least one second extension, the second aperture defining a second aperture longitudinal axis that is non-parallel with respect to the

first aperture longitudinal axis, each second arm translatable in connection with the second block along the second longitudinal axis, each first arm translatable in connection with the transverse extension along the first aperture longitudinal axis, at least one first arm translatable independently of translation of at least one second arm, and at least one second arm translatable independently of translation of at least one first arm.

[0012] In a yet further embodiment of the disclosure, a surgical retractor for retracting body tissue in a therapeutic procedure comprises at least one first arm defining proximal and distal ends and a first arm longitudinal axis extending therebetween, the distal end configured to support a retractor blade, the proximal end forming one of an extension or a block having an aperture; at least one second arm defining proximal and distal ends and a second arm longitudinal axis extending therebetween, the distal end configured to support a retractor blade, the proximal end forming one of an extension or a block having an aperture; a core: (a) configured to form a slideable connection each with the at least one first arm and the at least one second arm, (b) the slideable connection with the at least one first arm formed as a block having an aperture when the proximal end of the at least one first arm forms an extension, the slideable connection formed as an extension when the proximal end of the at least one first arm forms a block having an aperture, (c) the slideable connection with the at least one second arm formed as a block having an aperture when the proximal end of the at least one second arm forms an extension, the slideable connection formed as an extension when the proximal end of the at least one second arm forms a block having an aperture, (d) each first arm translatable using the slideable connection along an axis transverse to the first longitudinal axis, at least one first arm translatable independently of translation of at least one second arm, (e) each second arm translatable using the slideable connection along the second longitudinal axis, at least one second arm translatable independently of translation of at least one first arm, and (f) the distal end of at least one first arm and the distal end of at least one second arm mutually separable using the slideable connection to thereby separate retractor blades supported upon the at least one first arm and the at least one second arm.

[0013] In accordance with some exemplary embodiments, a retractor system can be provided, comprising a first arm having a first retractor blade configured to displace tissue, a second arm having a second retractor blade configured to displace tissue, and an engagement mechanism for engaging the first blade with the second blade to prevent movement between the first and second blades when the retractor system is in a non-extended position.

[0014] The retractor system can further comprise a third arm configured to support a posterior blade configured to displace tissue. The first arm, second arm and third arm can each be configured to be linearly translatable to be moveable with respect to each other. The engagement mechanism can be configured to lock the first blade with the second blade to prevent movement between the first and second blades. The engagement mechanism can comprise a tab disposed along a length of the first blade, and a slot configured along a corresponding length of the second blade configured to receive the tab.

[0015] The retractor system can further comprise a first docking pin sleeve provided in the first blade configured to receive a docking pin, and a first docking pin configured for placement in the first docking pin sleeve and configured to be threaded into a vertebral body of a patient to secure the retractor system.

[0016] The retractor system can further comprise a lengthening shim provided in the first blade configured to extend past a lower tip of the first blade to prevent tissue creep into the first blade. The retractor system can further comprise one or more detent stops provided along fixed, incremental locations along a length of the first blade. The retractor system can further comprise a lengthening shim detent tab configured to engage with one or more detent stops to allow for an adjustable height of the lengthening shim along the fixed, incremental locations. The one or more detent stops can be provided in 2.5 millimeter intervals. The retractor system can further comprise a lengthening shim removal hole on the lengthening shim configured to disengage the lengthening

shim detent tab from the one or more detent stops using a removal tool.

[0017] The retractor system can further comprise a widening shim provided along a lower portion of the first blade configured to extend past a lower tip of the first blade in a direction towards the second blade to prevent tissue creep. The retractor system can further comprise one or more detent stops provided along fixed, incremental locations along a width of the first blade. The retractor system can further comprise a widening shim detent tab configured to engage with the one or more detent stops to allow the widening shim to extend past the first blade along the fixed, incremental locations. The one or more detent stops can be provided in 1 millimeter intervals. The retractor system can further comprise a widening shim removal hole on the widening shim configured to disengage the widening shim detent tab from the one or more detent stops using a removal tool. The retractor system can further comprise one or more etched areas along a blade tip of the first blade for neuromonitoring.

[0018] The retractor system can further comprise a third arm having a posterior blade configured to displace tissue, and a disc shim secured to the posterior blade extending below the third blade to anchor the posterior blade into a disc space.

[0019] In some exemplary embodiments, a retractor system can be provided, comprising a first arm located around a central bore of the retractor system supporting a first retractor blade configured to displace tissue and configured to be translated independently, a second arm located around the central bore in proximity to the first arm and supporting a second retractor blade configured to displace tissue and configured to be translated independently, a tab disposed on a fixed location along a length of the first blade, and a slot disposed on a fixed location along a length of the second blade corresponding to the tab and configured to receive the tab to engage the first blade with the second blade and prevent movement of the first blade with respect to the second blade when the retractor system is in a non-retracted position.

[0020] In other exemplary embodiments, one or more of the retractor blades includes an opening extending from a first end to a second end of the blade. The opening is configured to receive a bone pin, which engages adjacent bone. The opening may be a partial channel in fluid communication with an interior of the blade. The channel may have a semi-circular cross-section and may extend along an entire length of the blade. In some embodiments, a portion of the channel may be threaded. In other embodiments, an insert, configured to receive a portion of the bone pin, may be positioned within the partial channel. The sleeve and/or pin may be at least partially contained within the blade, thereby moving the bone pin closer to the working channel for improved visibility while still keeping the bone pin out of the working window.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] A more complete understanding of the present invention, and the attendant advantages and features thereof, will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings wherein:

[0022] FIG. 1 depicts a surgical retractor blade holder of the disclosure, the blades held in a configuration conforming to a cannula;

[0023] FIG. 2 depicts the retractor of FIG. 1, the blades relatively separated in a manner operative to retract tissue of the body in a therapeutic procedure;

[0024] FIG. 3 depicts a retractor of the disclosure, including manually operable adjustment mechanisms for separating blades of the retractor, the blades in a closed configuration;

[0025] FIG. 4 depicts the retractor of FIG. 3, the blades in an opened configuration;

[0026] FIG. 5 depicts an alternative retractor of the disclosure, including adjustment mechanisms operable with a tool, the blades in a closed configuration;

[0027] FIG. 6 depicts the retractor of FIG. 5, the blades in an opened configuration;

[0028] FIG. 7 depicts an alternative retractor of the disclosure, including articulated blade holding arms, and a plurality of frame mounts of the disclosure;

[0029] FIG. 8 depicts a top view of the retractor of FIG. 7;

[0030] FIG. 9 depicts a top view of the retractor of FIG. 7, the blades in an opened position;

[0031] FIG. 10 depicts an alternative embodiment of a retractor of the disclosure, illustrating a thumbwheel operated rack and pinion adjustment mechanism;

[0032] FIG. 11 depicts a blade mounting configuration of the disclosure;

[0033] FIG. 12 depicts the blade mounting configuration of FIG. 11, connected to a retractor of the disclosure, the retractor configured with an alternative frame mount of the disclosure;

[0034] FIG. 13 depicts a light source retained by a blade holder of the disclosure;

[0035] FIG. 14 depicts an alternative blade type, and an alternative blade mounting configuration, of the disclosure;

[0036] FIG. 15 depicts an alternative blade pitch mechanism of the disclosure;

[0037] FIG. 16 depicts an articulated blade holding arm in accordance with the disclosure.

[0038] FIG. 17 is a diagram of the operative configuration of the retractor of FIGS. 1-2, in an open configuration;

[0039] FIG. 18 depicts the retractor of FIG. 17, in a closed configuration;

[0040] FIG. 19 is a diagram of an alternative retractor of the disclosure, in an open configuration;

[0041] FIG. 20 depicts the retractor of FIG. 19, in a closed configuration;

[0042] FIG. 21 is a diagram of another alternative retractor of the disclosure, in an open configuration;

[0043] FIG. 22 depicts the retractor of FIG. 21 in a closed configuration;

[0044] FIG. 23 depicts an alternative retractor system in a closed configuration in accordance with some embodiments;

[0045] FIG. 24 depicts the retractor system of FIG. 23 in an open configuration in accordance with some embodiments;

[0046] FIG. 25 depicts the retractor system of FIG. 23 in an open configuration with its posterior blade moved in accordance with some embodiments;

[0047] FIG. 26 depicts the retractor system of FIG. 23 in an open configuration with its cranial caudal blades moved in accordance with some embodiments;

[0048] FIG. 27 depicts the retractor system of FIG. 23 in an open configuration with its posterior and cranial caudal blades moved in accordance with some embodiments;

[0049] FIG. 28 depicts a top view of the retractor system of FIG. 23 in accordance with some embodiments;

[0050] FIG. 29 depicts a top view of the retractor system of FIG. 23 with the fixed plate removed in accordance with some embodiments;

[0051] FIG. 30 depicts a top perspective view of the retractor system of FIG. 23 with its components separated from one another in accordance with some embodiments;

[0052] FIG. 31 depicts a different top perspective view of the retractor system of FIG. 23 with its components separated from one another in accordance with some embodiments;

[0053] FIG. 32 depicts a top view of a portion of a three-blade retractor system in a closed or non-retracted position in accordance with some embodiments;

[0054] FIG. 33 depicts a top view of the retractor system of FIG. 32, with a portion removed to show mateable blade features in accordance with some embodiments;

[0055] FIG. 34 depicts a front view of a portion of a three-blade retractor system in a closed or a non-retracted position in accordance with some embodiments;

[0056] FIG. 35 depicts a top view of a portion of a two-blade retractor system in a closed or non-retracted position in accordance with some embodiments;

[0057] FIG. 36 depicts a top view of the retractor system of FIG. 35, with a portion removed to

show mateable blade features in accordance with some embodiments;

[0058] FIG. **37** depicts a front view of a portion of a two-blade retractor system in a closed or a non-retracted position in accordance with some embodiments;

[0059] FIG. **38** depicts a blade having a docking pin sleeve in accordance with some embodiments;

[0060] FIG. **39** depicts a cross-sectional view taken along line A-A of FIG. **38**;

[0061] FIG. **40** depicts a cross-sectional view taken along line B-B of FIG. **38**;

[0062] FIG. **41** depicts a docking pin inside a docking pin sleeve in accordance with some embodiments;

[0063] FIG. **42** depicts a blade having a lengthening shim in accordance with some embodiments;

[0064] FIG. **43** depicts a cross-sectional view taken along line A-A of FIG. **42**;

[0065] FIG. **44** depicts a cross-sectional view taken along line B-B of FIG. **42**;

[0066] FIG. **45** depicts a blade having a widening shim in accordance with some embodiments;

[0067] FIG. **46** depicts a top perspective view of the blade having the widening shim of FIG. **45**;

[0068] FIG. **47** depicts a top view of a portion of a retractor system including a blade having a widening shim in accordance with some embodiments;

[0069] FIG. **48** depicts a side view of a portion of a retractor system including a blade having a widening shim in accordance with some embodiments;

[0070] FIG. **49** depicts a perspective view of a portion of a blade including a disc shim in accordance with some embodiments;

[0071] FIG. **50** depicts a view of a blade separated from a disc shim in accordance with some embodiments;

[0072] FIG. **51** depicts a front view of a blade having a disc shim in accordance with some embodiments;

[0073] FIG. **52** depicts an alternative retractor system in a closed configuration in accordance with some embodiments;

[0074] FIGS. **53A-53D** shows several alternative views of a retractor blade according to one embodiment;

[0075] FIGS. **54A-54C** show alternative views of an anterior to psoas approach to the spine using the retractor system according to one embodiment;

[0076] FIGS. **55A-55B** show alternative views of a retractor blade with a sleeve for a bone pin according to another embodiment;

[0077] FIGS. **56A-56C** depict an alternative embodiment of a retractor blade with an insert for the bone pin according to another embodiment; and

[0078] FIGS. **57A-57B** depict an alternative embodiment of a retractor blade with an insert for the bone pin according to yet another embodiment.

DETAILED DESCRIPTION OF THE INVENTION

[0079] As required, detailed embodiments are disclosed herein; however, it is to be understood that the disclosed embodiments are merely examples and that the systems and methods described below can be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present subject matter in virtually any appropriately detailed structure and function. Further, the terms and phrases used herein are not intended to be limiting, but rather, to provide an understandable description of the concepts.

[0080] The terms “a” or “an”, as used herein, are defined as one or more than one. The term plurality, as used herein, is defined as two or more than two. The term another, as used herein, is defined as at least a second or more. The terms “including” and “having,” as used herein, are defined as comprising (i.e., open language). The term “coupled,” as used herein, is defined as “connected,” although not necessarily directly, and not necessarily mechanically.

[0081] FIGS. **1-2** illustrate a retractor system **10** that can be used to retract a patient's body tissue in

a surgical procedure in accordance with one embodiment of the present disclosure. Retractor system **10** includes a plurality of retractor blades **300**, and in this embodiment, a first blade **312**, a second blade **314**, and a third blade **316**. Blades **300** are each coupled to a retractor frame **100**, which includes first, second, and third arms **120**, **140**, and **160**, each having a blade holder **200** for holding and positioning a blade **300**. In one use of this embodiment, arms **120** and **140** are cranial/caudal blades, and blade **160** is a posterior blade, although other orientations and uses are contemplated. In an embodiment, each blade may be translated or indexed freely without a requirement to index another blade. In the embodiment shown, arms **120** and **140** are laterally located, and arm **160** is centrally located, although other relative dispositions are contemplated within the disclosure.

[0082] Retractor **10** is configured to be adjusted into a desired position, and then releasably fixed to an operating table or other object in the operating theatre, so that a relative position of retractor **10** and a patient can be controlled. Fixation of one or more of retractor **10** can be accomplished using one or more of an operating room (OR) table clamp, a retractor table arm, an arm clamp, and a frame clamp, for example.

[0083] Arms **120**, **140**, and **160** are each linearly translatable to be moveable with respect to each other. More particularly, arms **120** and **140** each include a base **150** having a slot **152**, sized to mateably receive and be linearly slideable upon an extension **154**. In an embodiment, arms **120** and **140** are mirror images of each other, although this is not necessary in order to carry out all aspects of the disclosure. In the embodiment illustrated in FIGS. 1-2, extension **154** has a flattened profile, thereby preventing unintended rotation of arms **120**, **140** with respect to each other, and as described below, with respect to arm **160**. Other mechanical forms can be provided for extension **154**, including for example a cylinder with a key, to enable a like purpose. In an alternative embodiment of the disclosure, any of arms **120**, **140**, and **160** can be rotated with respect to other of arms **120**, **140**, and **160**, where extension **154** or **170** (described below) is non-keyed, for example cylindrical.

[0084] A threaded set screw (not shown), ratchet (shown in other embodiments herein), or other mechanism can be provided to retain arm **120**, **140**, **160** in a desired linearly translated position. A rack and pinion configuration, not shown in this embodiment, can be provided, formed between extension **154** providing the rack, and with a pinion rotatably free but displaceably confined within arm **120/140** and engaged with the rack. A locking mechanism, for example a set screw or a ratchet mechanism, can be provided to affix extension **154**, **170** at a desired displacement.

[0085] Arm **160**, in this embodiment, includes an arm extension **170** which translates within a slot **172** within a block **174**, block **174** connected to extension **154**, for example with screws, or by welding, adhesives, or any other known means. In this manner, an orientation of extension **154** maintains a desired or predetermined orientation with respect to slot **172**. Accordingly, as arms **120**, **140** are translated upon extension **154**, or extension segments **154** that are mutually connected, and as arm **160** is translated within block **174**, a relative path of arms **120**, **140**, and **160** is maintained. Further, each arm **120**, **140**, **160** is independently translatable with respect to the other arms. In this manner, a medical practitioner can retract tissue in a manner which best suits a therapeutic purpose. Arm extension **170** can be provided with an extension stop **178**, to prevent an undesired separation of arm **160** from a remainder of retainer **10**.

[0086] As can be seen in FIGS. 1-2, arms **120**, **140** are sized and dimensioned to fit underneath block **174** when arms **120**, **140** are approximated. In this manner, an overall dimension of retractor **10** is minimized, thereby reducing a physical obstruction introduced by retractor **10**, and maximizing visualization within the surgical site. Arm **160** can be adjusted and affixed in any translated position as described with respect to arms **120**, **160** upon extension **154**, however a rack would be formed upon extension **170**, and a pinion disposed within block **174**.

[0087] It should be understood that in this embodiment, as well as other embodiments herein, while arms **120**, **140** are configured to form a slot, and translate upon an extension, they could be

configured to include extensions which translate within a block, as shown for arm **160**. Similarly, arm **160** could form a block which slides upon an extension, as shown for arms **120**, **140**. In one aspect of the disclosure, the relative configuration of arms **120**, **140** and arm **160** provides for an optimal packaging and a reduced physical profile, although various permutations of blocks and extensions as described herein can be provided, which can produce a reduced profile in a similar manner.

[0088] As can be seen in FIG. **1**, blades **300** are aligned relative to each other to form a tube **320**. In an embodiment, tube **320** has a tapered end **322**, and can function as a cannula. When retractor **100** is deployed, tube **320** can be formed by inserting blades **300** into an incision or opening in a patient's body, or can be positioned cooperative with other cannulae, to progressively increase a size of an opening in a body sufficient to admit passage of tube **320** with a minimum of disturbance to body tissue. As body tissue relaxes, or when desired, blades **300** can be separated or moved apart by moving arms **120** and **140** relatively apart. The relative extent of movement of arms **120**, **140** do not have to be uniform, and at different times, enabling the medical practitioner to control a timing and extent of force exerted upon different portions of body tissue relative to each blade **312**, **314**. Similarly, blade **316**, in connection with arm **160**, can be moved before, during, or after movement of either of arms **120**, **140**, to be closer or further from blades **312**, **314**.

[0089] As blades **312**, **314**, and **316** are moved relatively apart, a perimeter defined by an exterior surface of blades **312**, **314**, and **316** is increased, moving body tissue apart, and increasing access to an area within the body. A greater or fewer number of arms **120**, **140**, and **160** can be provided upon retractor **10** in a like configuration as arms **120**, **140**, or **160**, each configured to move independently of all other arms.

[0090] As can be seen in FIG. **2**, blades **312**, **314**, and **316** can be pivoted to be angled or pitched with respect to their respective arms **120**, **140**, **160**. In this manner, distal ends **330** of blades **300** can be further separated, and the perimeter defined by the exterior surface of the blades can be enlarged, without a necessity of increasing an overall profile of retractor **10**, or moving arms **120**, **140**, **160** any further apart.

[0091] FIG. **1** depicts retractor **10** in a “closed” or non-retracted configuration, in accordance with one embodiment of the present disclosure. In the closed configuration, blades **312**, **314**, and **316** are radially disposed around a central bore **324** to form the substantially closed, tube-shaped structure **320**. FIG. **2** depicts retractor system **10** in an “open” or retracted configuration, in which blades **312**, **314**, **316** have been pitched by being pivoted about a pivot **210** connected to a proximal end **332** of a blade **300**, and to an arm **120**, **140**, or **160**. In the open configuration, blades **300** no longer form a tube-shaped structure that is substantially closed.

[0092] Blades **300** can each be independently pitched or translated, and can be independently pitched or translated with respect to other blades. A stop element **212**, in the embodiment shown, a flange, extends from blade holder **200** and contacts arm **120**, **140**, or **160** at a desired extreme range of pivoting motion of blade holder **200** and associated blade **300**. A screw (not shown) can be provided within aperture **214**, which may be threaded, the screw configured to bear upon arm **120**, **140**, or **160** to cause pivoting of blade holder **200**, or to function as a stop element operative to change a maximum range of pivoting motion. Alternative embodiments are described with respect to FIGS. **5-6** and **15**, elsewhere herein.

[0093] Blade holder **200** is provided with a blade engagement profile **240** extending between opposing ends of blade holder **200**. Blades **300** are provided with a blade holder profile **340** mateable with blade engagement profile **240**, whereby when profiles **240**, **340** are mated, blade **300** can be retained upon blade holder **200** and be slideable along a length of blade holder profile **340** so that a penetration of blade **300** with respect to a patient is adjustable. Further, blades **300** can be replaced with blades having a different shape, size, or tissue engaging profile. In one embodiment, blade holder profile **340** and blade engagement profile **240** form a dovetail connection.

[0094] Additionally, blades **300** can be inserted after retractor **10** is fixed in a position with respect

to a patient, whereby a blade can be slid upwards and away from the patient, and replaced, without a requirement to move retractor **10** or the patient. In this embodiment, blade holder profile extends to a distal end **330** of blade **300**, or blade holder profile is configured to no longer provide a mating engagement along one or more portions of a length of blade **300**.

[0095] Blades **300** are affixed at a desired displacement along the length of blade holder profile **340** by a friction fit between blade holder profile **340** and blade engagement profile **240**, or by a set screw or other fastener (not shown) connected to blade holder **200** and contactable with blade **300** or blade holder profile **340**.

[0096] Blades **300** can be provided with one or more docking pin slots **350** sized and dimensioned to accept a docking pin (not shown), which can be used to secure a blade **300** in a position with respect to body tissue through which a pin is extended, for example into a vertebra. As all blades engaged upon retractor **10** are mutually connected, each docking pin thus inserted contributes to the overall stabilization of retractor **10**.

[0097] In an embodiment, retractor **10** is configured for an anterior approach to the spine of the patient. In this embodiment, shown in FIGS. **1-2**, arms **120**, **140** support caudal oriented blades, and arm **160** supports a cranial oriented blade. A range of pitch motion for blades **300** can include 0 degrees (forming tube **320**) to 20 degrees, although substantially larger angulation can be provided, for example 30, 40, or 50 degrees, or a greater range of angulation.

[0098] In surgical procedures where imaging is to take place, for example X-ray imaging, it is advantageous if at least blades **300** of retractor **10** are at least partially radiolucent, to foster visualization of the imaged area. Accordingly, blades **300** can be fabricated using aluminum, carbon fiber, or polymeric materials, or any other sufficiently strong and radiolucent material, or combination of materials, which is known or hereinafter developed.

[0099] With reference to FIGS. **3-6**, in an alternative frame embodiment **100A** of the disclosure, retractor **10A** includes elements similar to those of the embodiment of FIGS. **1-2**, and bears similar reference numbers. In FIGS. **3-4**, arms **120A**, **140A**, and **160A** support blades **300** in a manner similar to arms **120**, **140**, and **160** of FIGS. **1-2**, but arms **120A**, **140A** form square or rectangular channels which translate upon extensions **154A**. Similarly, arm **160A** translates within a rectangular channel forming block **174A**. Keys **180**, **182**, and **184** are positioned to be rotated by a medical practitioner while retractor **10A** is deployed, thereby translating arm **120A**, **140A**, or **160A**. With additional reference to FIGS. **5-6** and **11**, each of keys **180**, **182**, and **184** are connected to a pinion (not shown) rotatably fixed to arm **120A**, **140A** for keys **180**, **182**, and to block **174A** for key **184**. A toothed rack **186** is formed upon a side surface of each extension **154A**, and a similar toothed rack **188** is formed upon a side surface of arm **160**. In this manner, rotation of key **180**, **182**, or **184** causes a corresponding translational movement of arm **120A**, **140A**, and **160A**, respectively. In FIG. **4**, keys **180** and **182** have been rotated to separate arms **120A**, **140A**, and key **184** has been rotated to retract arm **160A**.

[0100] In FIGS. **5-6**, as can be seen in frame embodiment **100B**, in place of keys **180**, **182**, and **184**, tool engagements **190**, **192**, **194** are connected to rotatably fixed pinions mateable with racks **186**, **188**. Tool engagements **190**, **192**, and **194** are engageable by a driving tool, such as a hex or allen head driver. Accordingly, a driving tool (not shown) can be rotated by a hand of a medical practitioner, or by an electrical or computer controlled actuator, to adjust a position of arms **120B**, **140B**, or **160B**, and thereby change a position of one or more blades **300**, including for example blades **312**, **314**, or **316**. It should be understood that either a key **180** configuration, or a tool engagement **190** configuration, as shown in FIGS. **3-6**, can be provided in the embodiment of FIGS. **1-2**.

[0101] With further reference to FIGS. **5-6**, a blade pitch tool engagement **230**, **232**, **234** is provided, rotatable using a tool as described with respect to tool engagements **190**, **192**, and **194**, and operative thereby to change a pitch of a blade **300** attached to blade holder **200**.

[0102] More particularly, in one configuration, blade holder **200** is pivotally mounted at pivot **210**,

and blade pitch tool engagement **230**, **232**, and **234** are each rotatably retained upon their respective blade holder **200**. Threaded shaft **236** is keyed to prevent rotation, but is axially displaceable by rotation of blade pitch tool engagement **230**, **232**, **234**. As a result, shaft **236** can be caused to bear against arm **120B**, **140B**, **160B** to cause rotation of blade holder **200** about pivot **210**, and to thereby change an angle of blade **300**. In another embodiment, blade pitch tool engagement **230**, **232**, **234** is affixed to shaft **236**, and shaft **236** is threadably received within blade holder **200**. Thus, as blade pitch tool engagement **230**, **232**, **234** is rotated, shaft **236** bears against its respective arm **120B**, **140B**, **160B**. In a variation of this embodiment, shaft **236** is threadably received within arm **120B**, **140B**, **160B**. Other variations, including a threaded blade holder **200**, can be provided. In any of the foregoing embodiments, a biasing element (not shown) can be provided to bias blade holder **200** in a closed or pitched position, where a position of blade holder **200** is not positive controlled in each direction of rotation about pivot **210**. It should be noted that in one embodiment, the blade holder **200** may be positively controlled in each direction about the pivot **210**. While not shown for all embodiments, it should be understood that the foregoing blade holder pitch mechanism described for FIGS. **3-6** can be provided for other embodiments herein.

[0103] FIG. **15** illustrates an alternative embodiment, including frame **100J**, for controlling blade pitch. More particularly, changing an angulation or pitch of blade holder **200** is accomplished by rotating blade pitch tool engagement **230**, which has an attached threaded shaft **302**, within a corresponding threaded receiver **304** pivotally connected to blade holder at pin **306**.

[0104] Blade pitch tool engagement **230**, **232**, or **234** can be engaged with a tool, for example an allen or hex wrench, or may be provided with a knurled nut or manually engageable key. In this configuration, blade **300** can be pitched at a theoretically infinite range of angles. Other embodiments can be provided wherein blade **300** can be pitched at discrete stop points. A tool thus engaged can also be used to manipulate all or portions of retractor **10**, or a separate tool engagement can be provided upon retractor **10** to enable manipulation of other components of retractor **10** described herein.

[0105] With further reference to FIGS. **5-6**, a ratcheting pawl **250** can be provided, pivotable about pawl pivot **252**, a pawl **250** provided for any or all actuators having a toothed rack, including arms **120B**, **140B**, and **160B**, and all other embodiments herein upon which a rack is formed. A pawl tooth **254** has a sloped surface configured to allow rack **186**, **188** to pass in one direction, but not an opposite direction. In the embodiment shown, pawl **250** enables arms **120B**, **140B**, and **160B** to move apart, relative to each other. This operates to maintain tissue retracted during a therapeutic procedure. When the procedure is complete, or it is desired to otherwise remove retractor **10**, pawl **250** can be rotated about pivot **252** to disengage pawl tooth **254** from rack **186**, **188**, whereupon key **180**, **182**, and **184**, or tool engagement **190**, **192**, or **194** can be rotated to change a position of its associated arm.

[0106] In an alternative embodiment, pawl **250** does not include a sloped surface at pawl tooth **254**, and accordingly the pawl must be retracted from contact with rack **186** or **188** to enable movement of the associated arm.

[0107] FIGS. **5-6** further illustrate a frame mount **142**, which enables frame **100** (or frame **100A**, **100B**, and other frame embodiments herein) of retractor **10** to be mounted to a table or other supporting structure with a single mounting point. A support coupling **450** is affixed to the supporting structure with a rod (not shown) threadably or otherwise securable within an end of coupling **450**. Coupling **450** includes a threaded rod **148** and nut **148A** extending from an end portion. Coupling **450** is inserted into chamber **144** of frame mount **142**, and the attached threaded rod is passed into notch **146**. Nut **148A** is tightened to secure coupling **450** to frame mount **142**, and thereby securing retractor **10** to a supporting structure. The frame mount **142** illustrated includes two notches **146**, thereby enabling a support coupling to be connected to either side, or to both sides, of frame mount **142**. Two or more frame mounts **142** can be provided, as illustrated in FIGS. **7-9**, providing multiple or alternative mounting points for retractor **10**. In FIGS. **7-9**, it may

be seen that arm extension **160D** is connected to arm **160C** using a similar attachment mechanism. [0108] With further reference to FIGS. 5-6, a cam latch **220** is rotatably retained upon blade holder **200**, and is oriented to engage a slot (not shown) within blade **300**. In this manner, blade **300** is releasably retained upon blade holder **200**, and is prevented from sliding within blade holder profile **340**. Alternatively, cam latch **220** can overlap a protrusion (not shown) formed upon blade **300**.

[0109] The embodiment of FIGS. 7-9 is analogous to the embodiment of FIGS. 1-2 and 3-6, however arms **120C**, **140C** include arm extensions **120D**, **140D**, and arm couplings **430** and **432**. In one embodiment, rotation of coupling blade pitch tool engagements **430**, **432** are configured to change a pitch of blades **312**, **314** in a similar manner as described with respect to blade pitch tool engagements **230**, **232**. However, by positioning the corresponding pitch actuating components proximal with respect to blades **300**, less structure is imposed upon the target therapy site, improving visualization and manipulation within the target zone. Blade holders **200D** are configured to be compact, particularly as articulation does not need to be supported near a blade **300** attachment.

[0110] In an alternative embodiment, engagements **430**, **432** are coupled to pinions which rotate upon a rack formed in arm **120C** or **140C**. Thus, rotation of engagements **430**, **432** extend or retract arm extensions **120D**, **140D**. Alternatively, engagements **430**, **432** enable articulation of arm extensions **120D**, **140D** when rotated in one embodiment, and when loosed and manipulated in another embodiment. Articulation can be carried out by pivoting arm extension **120D**, **140D** about a pivot defined by engagement **430**, **432**, respectively. In a yet further embodiment, couplings **430**, **432** enable rapid replacement of arm extensions **120D**, **140D**. In a yet further embodiment, engagements **430**, **432** are connected to threaded shafts (not shown), which lie within elongated slots **328**. In this manner, arm extensions **120D**, **140D** can be displaced laterally with respect to arms **120C**, **140C**. In a variation of this embodiment, arm extensions **120D**, **140D** can be pivoted after lateral displacement within elongated slots **328**. An embodiment frame **100H** having elongated slots **328** disposed within arm **140D'** and arm extension **140H** is shown in FIG. 16.

[0111] With reference to FIG. 10, an embodiment of a retractor **10** of the disclosure includes a frame **100E** combining various elements described elsewhere herein, as indicated, and further includes thumbwheels **260** for changing a relative offset of arms **120E**, **140E**. In an embodiment, a pinion (not shown) rotatably connected to thumbwheel **260** engages rack **186** to cause movement of arm **120E** or **140E** relative to extension **154E**.

[0112] As can further be seen in FIG. 10, an additional moveable extension **170A** is provided with a rack **188A** and pinion (not shown) controllable, in this example, with a key **184A**. A mounting block **288** is connected to block **174**, whereby extension **170A** can be moved independently of extension **170**, and arm **188**. In an embodiment, extension **170A** is connected to a supporting structure, frame **100E** moveable with rotation of key **184A**.

[0113] With reference to FIGS. 11-12, a compact blade holder **200F** includes a mounting block **270** extending from or affixable to a remainder of retractor **10** using any known method. Block **270** includes a chamber **274** sized to receive a mounting holder extension **272**. A set screw **276** is threadably retained within holder extension **272**, and bears against a block flange **278** when extension **272** is inserted within chamber **274** and set screw **276** is tightened. Alternatively, set screw **276** can be threaded into block flange **278**. In this manner, mounting holder extension **272** is releasably securable within block **270**, and is thereby attachable to retainer **10**. A blade engagement profile **240** extends from holder extension **272**, and is connectable to a blade **300** as described with respect to FIGS. 1-2. In an alternative embodiment, blade **300** forms holder extension **272**, and therefore is directly attachable to block **270** as described above.

[0114] Referring now to FIG. 12, frame mount **142A** includes a central bore **280** through which a threaded screw **282** (not shown) can be passed, to threadably engage frame **100**. In this manner, frame mount **142A** can be rotated about an axis defined by bore **280** to a desired orientation. In this

embodiment, a key **284** is connected to screw **282**, and can be rotated to tighten screw **282** and thereby affix a position of frame mount **142A** relative to frame **100**.

[0115] In the embodiment of FIG. **12**, screw **282** threadably engages a slideable block that is retained within frame **100G** with a dovetail connection **292**, and is thereby longitudinally positionable within frame **100**. Extension **170G** of arm **160G** is slideably retained within slideable block **290**, and may be adjusted within block **290** using any of the methods disclosed elsewhere herein, including a rack and pinion configuration. In the embodiment shown, pinch bolts **294** can be rotated to tighten or loosen extension **170G**.

[0116] Referring now to FIG. **13**, a light source **500** can be affixed to blade **300**, or to blade holder **200**. Light source **500** can be provided with a blade holder profile **340** mateable with a blade engagement profile **240** of blade holder **200**. Alternatively, light source **500** can be affixed to blade holder **200** or blade **300** using any known means, including for example magnetic, hook and loop fastener, threadable faster, resilient attachment means, or adhesive. A target of focus of light emitted from light source **500** can be adjusted by changing a pitch of blade holder **200**, or by translating an associated arm **120**, **140**, **160**, as needed.

[0117] With reference to FIG. **14**, a variety of blade shapes **300** can be employed together with retractor **10** of the disclosure. In addition, while blades **300** can be provided with a blade holder profile, as illustrated in FIGS. **1-2**, blades **300** can alternatively be configured with a splined shaft **370**, which can be mateable with a corresponding receiver provided upon blade holder **200** to secure retain blade **300**, and to prevent rotation. In this manner, a wide variety of blade **300** styles and blade attachment mechanisms may be employed together with the various embodiments of the disclosure. In FIG. **14**, blades **300A** are configured to engage bone of the body, for example vertebrae. Blades **300A** include are angled at **372** to enable mounting blocks **200** to be positioned further from the target surgical zone, and are further provided with tapered ends **322A** which securely engage and retain bone. Accordingly, blades **300A** do not form a tube **320** in this embodiment. It may also be seen that blades **300/300A** can be provided in various lengths, widths, and if tubes **320** are formed, various lengths and diameter tubes. Blades **300/300A** can be provided in a variety of sizes, including for example 8 mm to 18 mm wide, and lengths from 110 mm to 200 mm, although substantially narrower, wider, shorter, or longer blades can be provided. Although two blade types are illustrated herein, a wide variety of blade types, styles, sizes and lengths can be used in accordance with the disclosure.

[0118] FIGS. **17-22** diagrammatically illustrate three possible embodiments for carrying out the disclosure, wherein heavy outline represents a block into which an extension is slideable. In FIGS. **17-18**, the embodiment of FIGS. **1-2** is depicted, wherein block **174** is affixed to extension **154**, thereby forming a fixed core. The remaining components, arms **120**, **140**, and **160**, are slideable independently of each other upon and within the core. In FIGS. **19-20**, extensions **154'** and **154''** extend from arms **120'** and **140'**. Extensions **154'**, **154''**, and **160** slide at different height offsets within block **174**, which alone forms the core within which the moveable arms slide. In FIGS. **21-22**, arms **120** and **140** form blocks, as in FIGS. **17-18**, and arm **160** is formed as a block **174'**, slideable upon extension **160'**, which is affixed to extension **154**, extensions **160'** and **154** forming a core upon which the moveable arms slide. It should be understood that only a single arm **120** or **140** can be provided, or that multiple arms **120** or **140**, or multiple arms **160** can be provided, as benefit the therapeutic needs of the patient.

[0119] Frame **100** and its constituent components (including variants **100A-100H**) may be fabricated from any one or more, or combinations of, metals, polymers, carbon fiber or other composites, natural materials, or any other materials having sufficient strength, durability, and biocompatibility. Materials selected can be radiolucent or radiopaque, as desired.

[0120] In accordance with the disclosure, a single mounting location for a frame **100** of the disclosure enables a posterior blade (e.g. **316**) to be indexed without moving the cranial caudal blades (e.g. **312**, **314**), as well as allowing the cranial caudal blades to be indexed without moving

the posterior blade.

[0121] FIGS. **23-31** depict different views of an alternative retractor system in accordance with some embodiments. The retractor system **610** includes a number of features as in prior embodiments, including first, second, and third blades **612, 614, 616**; first, second, and third arms **620, 640, 660**; and various engagement mechanisms for independently translating the various blades and arms. The retractor system **610** also includes a number of novel features including a carriage actuator **672** and a third (e.g., posterior) blade actuator **674** that provides different advantages during a surgical procedure, as discussed in more detail below.

[0122] FIG. **23** depicts an alternative retractor system in a first or closed configuration in accordance with some embodiments. The retractor system **610** comprises a frame including a first arm **620**, a second arm **640** and a third arm **660** for holding and attaching one or more blades thereto. A first blade **612** having a first blade arm **613** attaches to the first arm **620**. A second blade **614** having a second blade arm **615** attaches to the second arm **640**. And a third blade **616** having a third blade arm **617** attaches to the third arm **660**. The first blade arm **613** attaches to the first arm **620** via first fastener **621**. The second blade arm **615** attaches to the second arm **640** via second fastener **641**. The third blade arm **617** attaches to the third arm **660** via third fastener **661**. In some embodiments, the first and second blades **612, 614** can be considered cranial caudal blades in view of their direction of movement, while the third blade **616** can be considered a posterior blade.

[0123] The retractor system **610** is in a first or closed configuration whereby each of the blades **612, 614, 616** are adjacent one another. The retractor system **610** can advantageously be delivered into a patient in the closed configuration through an open, mini-open or minimally invasive incision. Once in the patient, each of the blades **612, 614, 616** is capable of angulation to thereby expand and retract tissue. Furthermore, each of the arms **620, 640, 660** (and their associated blades) are capable of movement (e.g., translational) in one or more directions, to thereby expand and retract tissue. In some embodiments, one or more blades **612, 614, 616** are angulated away from one another. In other embodiments, one or more blades **612, 614, 616** are translated away from one another. In other embodiments, one or more blades **612, 614, 616** are both angulated and translated away from one another. By separating the blades via angulation or translational movement, this creates a second or open configuration.

[0124] Each of the blades **612, 614, 616** is capable of angulation. Advantageously, each of the blades **612, 614, 616** has its own individual mechanism to provide angulation, such that each blade can be angulated independently from one another. A first blade pitch tool engagement or blade angle actuator **630** angulates first blade **612**. A second blade pitch tool engagement or blade angle actuator **632** angulates second blade **614**. A third blade pitch tool engagement or blade angle actuator **634** angulates third blade **616**. In some embodiments, the blade angle actuators **630, 632, 634** are angulated by hand and/or an instrument. In some embodiments, each of the blade angle actuators **630, 632, 634** resides within a base plate. In some embodiments, the base plates pivot around an axis, thereby accommodating the movement caused by the actuators **630, 632, 634**.

[0125] Each of the arms **620, 640, 660** is capable of movement, such as translational movement. First arm **620** is capable of linear movement via rotation of first tool engagement or linear actuator **690**. The first linear actuator **690** can be rotated by hand and/or instrument. As the first linear actuator **690** is rotated, the lower portion of the actuator **690** engages teeth **691** (shown in FIG. **24**) as part of a rack and pinion system, thereby advantageously causing the first arm **620** to linearly translate. In some embodiments, rotation of the first linear actuator **690** causes the first arm **620** to move laterally outward (e.g., away from a longitudinal axis that extends through the system frame), thereby separating the first arm **620** from the second arm **640** and third arm **660**. Rotation of the first linear actuator **690** in an opposite direction causes the first arm **620** to move laterally inward (e.g., toward a longitudinal axis that extends through the system frame), thereby bringing the first arm **620** closer to the second arm **640** and third arm **660**.

[0126] Likewise, second arm **640** is capable of linear movement via rotation of second tool

engagement or linear actuator **692**. The second linear actuator **692** can be rotated by hand and/or instrument. As the second linear actuator **692** is rotated, the lower portion of the actuator **692** engages teeth **693** (shown in FIG. **24**) as part of a rack and pinion system, thereby advantageously causing the second arm **640** to linearly translate. In some embodiments, rotation of the second linear actuator **692** causes the second arm **640** to translate or move laterally outward (e.g., away from a longitudinal axis that extends through the system frame), thereby separating the first arm **620** from the second arm **640** from the first arm **620** and third arm **660**. Rotation of the second linear actuator **692** in an opposite direction causes the second arm **640** to move laterally inward (e.g., toward a longitudinal axis that extends through the system frame), thereby bringing the second arm **640** closer to the first arm **620** and third arm **660**.

[0127] In addition to accommodating linear lateral translation of one or more arms (e.g., the first and second arms that attach to the first and second cranial caudal blades), the retractor system **610** can also provide linear translation of one or more arms in a longitudinal direction that runs along a length of the retractor system. In some embodiments, the retractor system **610** comprises a carriage actuator **672** that actuates a moveable carriage **670** to which the first arm **620** and the second arm **640** are attached, thereby causing the first blade **612** and the second blade **614** to translate in a longitudinal or anterior/posterior direction. In some embodiments, the carriage actuator **672** actuates the moveable carriage **670** via a rack and pinion system. In some embodiments, the retractor system **610** further comprises a posterior blade actuator **674** that causes translation of the third arm **660**, thereby causing the third blade **616** to translate in a longitudinal or anterior/posterior direction. In some embodiments, the posterior blade actuator **674** actuates the third arm **660** via a rack and pinion system.

[0128] In some embodiments, a fixed plate **680** is provided with a moveable carriage or carriage plate **670** positioned thereunder. As shown in FIG. **23**, fixed plate **680** comprises an opening for receiving the third arm **660** therethrough. In some embodiments, the third arm **660** is in the form of a dovetail, whereby it forms a dovetail connection with the fixed plate **680**. A moveable carriage **670** is positioned on an underside of the fixed plate **680**, and is configured to translate while holding the first blade **612** and the second blade **614** via rotation of the carriage actuator **672**. On an upper surface of the fixed plate **680**, carriage actuator **672** and posterior blade actuator **674** extend therefrom.

[0129] Rotation of the carriage actuator **672** causes the carriage **670** to move or translate linearly in a longitudinal or posterior/anterior direction. In some embodiments, the fixed plate **680** remains in place, while the carriage **670** translates relative to the fixed plate **680**. As the carriage **670** is attached to both the first arm **620** and the second arm **640**, while the first arm **620** is attached to the first blade **612** and the second arm **640** is attached to the second blade **614**, movement of the carriage **670** causes translational movement of both the first blade **612** and the second blade **614**. Rotation of the carriage actuator **672** in a first direction causes the first and second blades (e.g., the cranial caudal blades) **612**, **614** to move away from the third blade (e.g., posterior blade) **616**. Rotation of the carriage actuator **672** in a second direction causes the first and second blades (e.g., the cranial caudal blades) **612**, **614** to move toward the third blade (e.g., posterior blade) **616**.

[0130] Rotation of the posterior blade actuator **674** causes the third arm **660** to move or translate linearly in a longitudinal or posterior/anterior direction. As the third arm **660** is attached to the third blade (e.g., the posterior blade) **616**, movement of the posterior blade actuator **674** causes translational movement of the third blade **616**. Rotation of the posterior blade actuator **674** in a first direction causes the third blade **616** to move away from the first and second blades (e.g., the cranial caudal blades) **612**, **614**. Rotation of the posterior blade actuator **674** in a second direction causes the third blade (e.g., the posterior blade) to move toward the first and second blades (e.g., the cranial caudal blades) **612**, **614**.

[0131] The carriage actuator **672** and the posterior blade **674** provide unique advantages to the present system. For example, a surgeon may position the retractor system **610** such that the third

blade **616** is next to or blocking a nerve (e.g., a femoral nerve), but may want to retract tissue further in a longitudinal direction. Rather than having to move the entire retractor system **610**, the present retractor system **610** simply allows the surgeon to rotate the carriage actuator **672**, thereby providing translation of the first and second blades **612**, **614**, without having to move the entire system. In another example, a surgeon may have positioned the retractor system **610** such that the first and second blades **612**, **614** are in a desired position, while the third blade **616** has room to be brought closer to a nerve (e.g., a femoral nerve). Rather than having to move the entire retractor system **610**, the present retractor system **610** simply allows the surgeon to rotate the posterior blade actuator **674**, thereby providing translation of the third blade **616** without having to move the entire system, thus advantageously increasing retractor accuracy and saving time during a surgical procedure.

[0132] In some embodiments, the retractor system **610** further includes a first attachment mechanism **696** and a second attachment mechanism **698** for attaching additional components to the frame. Each of the attachment mechanisms **696**, **698** extends from an upper surface of the retractor system **610**. The attachment mechanism **696**, **698** comprise extensions or protrusions for which a component can be attached thereon. As shown in FIG. **23**, the first attachment mechanism **696** is positioned closer to the blades **612**, **614**, **616** than the second attachment mechanism **698**. In some embodiments, a fourth arm and/or blade can be attached to the first attachment mechanism **696**. In some embodiments, a table mount component can be attached to the second attachment mechanism **698**, such that the entire frame with the blades can be attached to a table.

[0133] FIG. **24** depicts the retractor system of FIG. **23** in a second or an open configuration in accordance with some embodiments. As shown in the figure, the first blade **612**, second blade **614** and third blade **616** have been separated apart from one another. In some embodiments, the first linear actuator **690** has been rotated to move or translate the first blade **612** away from the second blade **614** and the third blade **616**. Likewise, the second linear actuator **692** has been rotated to move or translate the second blade **614** away from the first blade **612** and the third blade **616**. In the open configuration in FIG. **23**, the blades provide a greater channel through which one or more instruments and/or implants can be inserted through the retractor system **610**.

[0134] FIG. **25** depicts the retractor system of FIG. **23** in a different second or open configuration with its posterior blade moved in accordance with some embodiments. Like the retractor system **610** in the previous figure, the retractor system **610** is in an open configuration whereby the first blade **612**, second blade **614** and third blade **616** are separated from one another. The first blade **612** has been moved away from the second blade **614** and the third blade **616** via the first linear actuator **690**, while the second blade **614** has been moved away from the first blade **612** and the third blade **616** via the second linear actuator **692**. In addition, as shown in FIG. **25**, the third blade **616** (e.g., the posterior blade) has been moved linearly away from the first and second blades **612**, **614** (e.g., the cranial caudal blades) via posterior blade actuator **674**. As shown in the figure, upon rotation of the posterior blade actuator **674**, the third arm **660** moves back through the fixed plate **680** of the frame of the retractor system **610**, thereby exposing a posterior portion **661** of the third arm **660**.

[0135] FIG. **26** depicts the retractor system of FIG. **23** in a different second or open configuration with its cranial caudal blades moved in accordance with some embodiments. Like the retractor system in FIG. **24**, the retractor system **610** is in an open configuration whereby the first blade **612**, second blade **614** and third blade **616** are separated from one another. As shown in FIG. **26**, the carriage actuator **672** has been rotated such that the carriage **670** attached to the first blade **612** and the second blade **614** has been translated linearly (e.g., in an anterior direction). In other words, the first and second blades **612**, **614** (e.g., the cranial caudal blades) have been shifted anteriorly in accordance with some embodiments. From FIG. **26**, one can see how the carriage **670** has been shifted relative to the fixed plate **680**. In addition, the first blade **612**, second blade **614** and third blade **616** have been angulated via blade angle actuators **630**, **632** and **634**.

[0136] FIG. 27 depicts the retractor system of FIG. 23 in a different second or open configuration with its posterior and cranial caudal blades moved in accordance with some embodiments. Like the retractor system in FIG. 24, the retractor system 610 is in an open configuration whereby the first blade 612, second blade 614 and third blade 616 are separated from one another. As shown in FIG. 27, the carriage actuator 672 has been rotated such that the carriage 670 attached to the first blade 612 and the second blade 614 has been translated linearly (e.g., in an anterior direction). In other words, the first and second blades 612, 614 (e.g., the cranial caudal blades) have been shifted anteriorly in accordance with some embodiments. From FIG. 27, one can see how the carriage 670 has been shifted relative to the fixed plate 680. In addition, the posterior blade actuator 674 has been rotated such that the third blade 616 (e.g., the posterior blade) has been moved linearly away from (e.g., in a posterior direction) from the first blade 612 and the second blade 614. From FIG. 27, one can see how the third arm 660 has been translated linearly such that at least a posterior portion 661 of it is now exposed from the fixed plate 680. In addition, the first blade 612, second blade 614 and third blade 616 have been angulated via blade angle actuators 630, 632 and 634. As shown by each of the configurations in FIGS. 24-27, the blades can have various spacings and angles relative to one another.

[0137] FIGS. 28 and 29 depict top views of the retractor system of FIG. 23 in accordance with some embodiments. FIG. 28 shows the retractor system 610 with the fixed plate 680, while FIG. 29 shows the retractor system 610 with the fixed plate 680 removed. From these views, one can see how the various actuators and knobs actuate and translate the arms and blades of the retractor system. For example, one can see how rotation of the first linear actuator 690 causes a bottom gear portion of the actuator 690 to engage teeth 691 (shown in full view in FIG. 29), thereby causing the first arm 620 and first blade 612 to linearly translate. Rotation of the second linear actuator 692 likewise engages a different set of teeth 693 (shown in full view in FIG. 30), thereby causing the second arm 640 and second blade 614 to linearly translate.

[0138] FIGS. 30 and 31 depict different top perspective views of the retractor system of FIG. 23 with its components separated from one another in accordance with some embodiments. In particular, there are two distinct portions—a blade portion and a frame portion. The blade portion includes the blades 612, 614, 616 and their respective arms 613, 615, 617, while the frame portion includes the fixed plate 680, carriage 670 and all actuator knobs. From these views, one can see how the blade portion is attached to the frame portion via first, second and third fasteners 621, 641, 661 that are received through corresponding openings 623, 643, 663 formed in the frame portion.

[0139] Below are additional advantageous features that can be used with any of the retractor systems discussed above. FIGS. 32-37 illustrate blade engagement mechanisms that allow one or more blades to interlock, engage or attach to another. These engagement mechanisms, which can be in the form of one or more mating tabs, advantageously help to reduce movement between the blades, particularly when they are in a closed configuration. FIGS. 38-41 illustrate retractor blades including one or more inner sleeves for receiving a docking element, such as a docking pin. The docking pin can advantageously help to secure the blade to a vertebra, thereby helping to stabilize the blade during surgery. FIGS. 42-46 show a slidable lengthening shim that can be slid down a blade, thereby extending the tip of the blade. The lengthening shim advantageously serves to increase the tip of the blade to prevent tissue creep from occurring. FIGS. 47 and 48 show a slidable widening shim that can be slid down a blade, thereby extending the width of the blade. The widening shim advantageously serves to increase the width of the blade to prevent tissue creep from occurring. FIGS. 49-51 show a slidable disc shim that can be slid down a blade. The disc shim advantageously allows a blade to anchor into a disc space, thereby providing greater stabilization for the retractor systems and reducing the likelihood of the retractor system moving from a target site. These features are discussed in more detail below.

[0140] FIGS. 32-37 illustrate blade engagement mechanisms that allow one or more blades to interlock, engage or attach to another. FIG. 32 illustrates a top view of a portion of a retractor

system **700** that can be used to retract a patient's body tissue in a surgical procedure in accordance with one embodiment of the present disclosure. From this view, one can see blade arms that attach to respective arms of a frame, such as first blade arm **710**, second blade arm **720** and a third blade arm **730**. The first blade arm **710** can attach to a first arm of the frame via first fastener **721**. The second blade arm **720** can attach to a second arm of the frame via second fastener **741**. The third blade arm **730** can attach to a third arm of the frame via third fastener **761**.

[0141] In some embodiments, the first blade arm **710** can have a first blade **712** disposed within the first blade arm **710**, the second blade arm **720** can have a second blade **722** disposed within the second arm **720**, and the third blade arm **730** can have a third blade **732** disposed within the third blade arm **730**. In some embodiments, the first and second blades **712**, **722** can be deemed cranial/caudal blades, while the third blade **732** can be deemed a posterior blade, although other orientations and uses are contemplated. In some embodiments, each blade may be translated or indexed freely without a requirement to index another blade. In an embodiment shown, blade arms **710** and **720** are laterally located, and blade arm **730** is centrally located relative to the other blades arms, although other relative dispositions are contemplated within the present disclosure.

[0142] In some embodiments, the retractor system **700** can be configured to be adjusted into a desired position, and then releasably fixed to an operating table or other object in an operating room. Fixation of one or more portions of a retractor system **700** can be accomplished using, for example, one or more of an operating room table clamp, a retractor table arm, an arm clamp, a frame clamp or other similar device. Blade arms **710**, **720** and **730** can each be linearly translatable to be moveable with respect to each other. In some embodiments, blade arms **710** and **720** can be mirror images of each other, although this is not required.

[0143] In some embodiments, one or more blades **712** and **722** can have an engagement mechanism **750** to engage one blade with another during installation of the retractor system or during a procedure. In some embodiments, at least one blade comprises a mateable tab, prong or extension that fits into an opening, recess or slot in another blade, thereby interlocking the blades. For example, blade **712** can have a tab **752** that can extend along a length of the blade **712**, and blade **722** can have a slot **754** that can receive the tab **752** at a corresponding length of the blade **722**. In some embodiments, the tab **752** can be integral to the blade **712** or a separate component. For example, in some embodiments, the tab **752** can be formed monolithically with the blade **712**, or can be welded to the blade **712**. In some embodiments, the slot **754** can be configured to frictionally engage with the tab **752** so that the tab **752** is fittingly received by the slot **754**. The engagement mechanism **750** can stabilize the retractor system **700** by keeping the blade **712** engaged with the blade **722**, and can result in stability between the arm **710** and arm **720** such as during installation of the retractor system **700** or during a procedure.

[0144] Various engagement mechanisms are contemplated by the present disclosure and are not limited to the tab **752** and slot **754** mechanisms described herein. The engagement mechanism **750** can engage, mate or lock one blade with the other, preventing movement of one blade with respect to the other during, for example, a procedure or installation of the retractor. In some exemplary embodiments, other engagement mechanisms can be used, such as a slot and a pin, a screw in a threaded hole, magnets, or other mechanisms allowing the blade **712** to be engaged or secured to the blade **722**. The engagement mechanism can be integral with the blades or a separate component.

[0145] FIGS. **32-34** illustrate retractor system **700** in a “closed” or non-retracted configuration, in accordance with an embodiment of the present disclosure. In the closed configuration, first blade **712**, second blade **722** and third blade **732** are radially disposed around a central bore **740** to form a substantially closed, tube-shaped structure. The blades can each be independently pitched or translated, and can be independently pitched or translated with respect to other blades.

[0146] In some exemplary embodiments, as shown in FIGS. **35-37**, a retractor system **800** can be provided with a two-blade configuration. The retractor system **800** can comprise a first blade **812**

that extends from a first blade arm **810** and a second blade **822** that extends from a second blade arm **820**. In some embodiments, blades **812** and **822** can have an engagement mechanism **830** to prevent movement of one blade with respect to the other during installation of the retractor or during a procedure. For example, blade **812** can have a tab **832** that can extend along a length of the blade **812**, and blade **822** can have a slot **834** that can receive the tab **832** at a corresponding length of the blade **822**. In some embodiments, the slot **834** can be configured to frictionally engage with the tab **832** so that the tab **832** is fittingly received by the slot **834**.

[0147] FIGS. **38-41** illustrate retractor blades including one or more inner sleeves for receiving a docking element, such as a docking pin. In some embodiments, a retractor system **900** comprises one or more blades **910** including a docking pin sleeve **920** provided therein. The docking pin sleeve **920** can be provided in an interior wall of the associated blade **910**. In some embodiments, the docking pin sleeve **920** can be received within a slot **930** formed along an inner wall of the blade **910**. In some embodiments, the slot **930** can allow for various instruments to slide down the blade **910** if desired. In some embodiments, a docking pin is received along the slot **930** and into the docking pin sleeve **920**. The docking pin can advantageously engage bone, thereby helping to stabilize the associated retractor blade in preparation for or during a surgical procedure.

[0148] As shown in FIG. **39**, which is a cross-sectional view of the line A-A shown in FIG. **38**, in some embodiments, the docking pin sleeve **920** can be a hollow shaft within the blade **910** that allows for placement of a docking pin. In some embodiments, the docking pin sleeve **920** is a hollow cylindrical body, while in other embodiments, it can be other shapes, such as square or rectangular. As shown in FIG. **40**, which is a cross-sectional view of the line B-B shown in FIG. **38**, in some embodiments, one or more holes **926** are provided within the walls **922**, **924** of the docking pin sleeve **920** to accommodate a docking pin.

[0149] FIG. **41** depicts a docking pin inside a docking pin sleeve in accordance with some embodiments. The docking pin **950**, which can comprise a head portion and a threaded shaft, can be inserted through the docking pin sleeve **920**. The docking pin **950** can slide down the docking pin sleeve **920** and can be threaded into the vertebral body to anchor the retractor blade to a bone.

[0150] FIGS. **42-46** show a slidable lengthening shim that can be slid down a blade, thereby extending the tip of the blade. The lengthening shim advantageously serves to increase the tip of the blade to prevent tissue creep from occurring. In some exemplary embodiments, as shown in FIGS. **42-46**, a lengthening shim can be provided within a blade. For example, a blade **1010** can extend from a blade arm **1000**, which will be attached to a retractor system frame.

[0151] A slot **1020** can be provided within the blade **1010** for instruments to slide down the slot for use in various procedures during and after installation of the retractor system. In some embodiments, a lengthening shim **1030** can be provided within the blade **1010**. A lengthening shim detent tab **1040** (shown in FIG. **44**) can be provided on the lengthening shim **1030** to adjust the height of the lengthening shim **1030** that extends out of the retractor arm **1000**. In some exemplary embodiments, one or more detent stops **1050** (shown in FIG. **44**) can be provided within the blade **1010** to engage with the lengthening shim detent tab **1040** to adjust the height of the lengthening shim **1030**. Any number of detent stops **1050** can be provided at any fixed, incremental locations, and the present disclosure is not limited to any particular number or distance between the detent stops **1050**. For example, in some exemplary embodiments, four detent stops **1050** can be provided in 2.5 millimeter increments where the top detent stop **1050** will keep the lengthening shim **1030** and blade **1010** flush with the bottom of the retractor arm, and the bottom detent stop **1050** will allow the lengthening shim **1030** and blade **1010** to extend below the retractor arm **1000** a distance of ten millimeters. In other words, the lengthening shim **1030** can be moved incrementally (e.g., in millimeters), with the distance travelled between the stops serving as the distance the lengthening shim **1030** can travel down the blade. The lengthening shim **1030** can advantageously incrementally increase the tip of the blade **1010** and can prevent tissue creep underneath the blade **1010**. A lengthening shim removal hole **1060** can be provided to disengage the lengthening shim

detent tab **1040** with the detent stop **1050**, using a removal tool.

[0152] FIGS. **47** and **48** show a slidable widening shim that can be slid down a blade, thereby extending the width of the blade. The widening shim advantageously serves to increase the width of the blade to prevent tissue creep from occurring. In some exemplary embodiments, as shown in FIGS. **47** and **48**, a widening shim can be provided within a blade. For example, a widening shim **1110** can be delivered down one or more blades. The widening shim prevents tissue from creeping in between adjacent blades, particularly when they are in an open configuration. The widening shim **1110** can be slidably provided down a slot **1120** formed within the blade **1100**. The widening shim can be any length, and in some embodiments, is between 1-2 inches. In some exemplary embodiments, one or more detent stops can be provided along a width of the blade **1100** to engage with one or more widening shim detent tabs to adjust the widening shim **1110**. Any number of detent stops can be provided at any fixed, incremental locations, and the present disclosure is not limited to any particular number or distance between the detent stops. For example, in some exemplary embodiments, ten detent stops can be provided in 1 millimeter increments to allow the widening shim **1110** to extend past the blade edge a total of ten millimeters. In some embodiments, the widening shim **1110** can advantageously prevent and block tissue creep underneath around the sides of the blade **1100**. A widening shim removal hole can be provided to disengage the widening shim detent tab with the detent stop, using a removal tool.

[0153] As shown in FIG. **47**, once a retractor **1160** is installed, the blades **1162**, **1164** can be moved from a closed to an open (or retracted) position. A gap **1170** can exist between adjacent blades **1162**, **1164**. A widening shim **1110** can be slid down a blade **1162**. The widening shim **1110** includes a wall that extends in between the gap **1170**, thereby preventing creep from occurring in the gap **1170**. In some embodiments, a widening shim can also be provided within the blade **1164** to extend towards the blade in arm **1162**. For example, when the retractor is opened a distance greater than the widening shim **1110** can extend in blade **1162**, a second widening shim in blade **1164** can be provided to extend past the blade **1162** toward the blade **1162**. In some exemplary embodiments, etched areas **1150** can be provided along the blade tip **1140** for neuromonitoring.

[0154] FIGS. **49-51** show a slidable disc shim that can be slid down a blade. In some embodiments, a disc shim can be provided to aid a posterior blade of a retractor to be anchored into a disc space. This can provide further stabilization of the retractor and a reduced chance of the retractor moving from an initial target. Referring to FIGS. **49-51**, for example, a disc shim **1210** can have an upper portion **1225** and a lower narrower portion **1215**. The lower narrower portion **1215** can be inserted within a slot **1230** in the posterior blade **1220**, and the upper portion can wrap around the edges of the posterior blade **1220** to keep it in place. As shown in FIG. **51**, as the disc shim **1210** slides down the posterior blade **1220**, into the disc, locking tabs **1260** can be provided on outer edges of the posterior blade **1220** to engage the upper portion **1225** of the disc shim **1210** and lock the disc shim **1210** in place once the disc shim **1210** is fully extended, and can prevent the disc shim **1210** from moving upwards. Further, stops **1270** can be provided along the outer edges of the posterior blade which can engage the upper portion **1225** of the disc shim **1210**, and can prevent the disc shim **1210** from advancing downward too far beyond the tip **1235** of the blade **1220** distally. The lower narrower portion **1215** can allow for the posterior blade **1220** to be anchored into the disc space. In some exemplary embodiments, the disc shim **1210** can have slots which can allow the edges of the disc shim **1210** to be displaced outwards by using a removal tool. The removal tool can be threaded into the disc shim **1210** to disengage the locking tabs **1260** allowing the disc shim **1210** to be easily removed.

[0155] FIGS. **52-54** depict different views of an alternative retractor system in accordance with some embodiments. The retractor system **1610** includes a number of features as in prior embodiments, including first, second, and third blades **1612**, **1614**, **1616**; first, second, and third arms **1620**, **1640**, **1660**; and various engagement mechanisms for independently translating and rotating the various blades and arms. The retractor system **1610** includes a frame body with a

number of different actuators, which has substantially the same functionality as described for system **610** discussed above and therefore will not be repeated.

[0156] FIGS. **53A-53D** show several alternative views of retractor blade **1612**; although it will be appreciated that the same retractor blade features may be provided on retractor blade **1614**, vice versa, or the like. Retractor blade **1612** may include one or more inner sleeves **1622** for receiving a docking element, such as a docking pin or bone pin **1650**. The sleeve **1622** can be a hollow shaft or channel within or through the blade **1612** that allows for placement of a docking pin, bone pin, or the like. In some embodiments, the sleeve **1622** is a hollow cylindrical opening, while in other embodiments, it can be other shapes, such as square or rectangular. The sleeve **1622** may extend from a first end to a second end of the associated blade **1612**, for example, from the proximal to distal-most end. In some embodiments, the docking pin sleeve **1622** can be received within a protrusion **1613** formed along an outer wall of the blade **1612** as best seen in FIGS. **53A** and **53B**. As seen in FIG. **53C**, a proximal portion of the sleeve **1622** can be threaded, for example, to engage with a threaded portion of the docking pin **1650**. Alternatively, the sleeve **1622** can be non-threaded or the sleeve **1622** may have threaded and non-threaded portions along its length.

[0157] The docking pin **1650** may comprise a head portion **1652** and a threaded shaft with a distal most tip **1654** that can be inserted through the docking pin sleeve **1622**. The docking pin **1650** can slide down through the docking pin sleeve **1622** and can be threaded into the vertebral body to anchor the retractor blade **1612** to the bone. One advantage of threading the pin **1650** into the blade **1612** is the mechanical advantage of when the pin **1650** is being driven into the bone, for example, when the bone is very dense or hard. The docking pin **1650** can advantageously engage the bone to help to stabilize the associated retractor blades in preparation for or during a surgical procedure. As best seen in FIG. **54C**, the position of the pins **1650** on the outside of the blades **1612**, **1614** allows for a clear working window. Unlike other designs where the pins **1650** may be inserted through a central hole blocking the surgeons viewing window, the blade design allows the surgeon more visibility in the viewing window. In addition, a light source may be inserted at the same time with the pin **1650** further enhancing visibility.

[0158] According to one method for accessing the spine depicted in FIGS. **54A-54C**, an anterior to psoas approach may be advantageous. In this approach, the surgeon does not need to dissect through the psoas muscle in order to access the disc space, which typically occurs in a lateral approach, or through the abdominal cavity, which typically occurs in an anterior approach. As shown in FIG. **54A**, arrow A depicts a generally anterior approach and arrow L depicts a generally lateral approach to the spine. The anterior to psoas approach, as depicted by the relative position of the retractor **1610**, is angled in an orientation between the arrow A and arrow L approaches. The anterior to psoas methodology is therefore less invasive and more muscle sparing. The retractor system **1610** shown in FIG. **52** may be especially suited for this anterior to psoas approach. The orientation of the retractor **1610** during the procedure is such that the blades **1612**, **1614**, **1616** are angled between a direct lateral and a direct anterior with one blade (blade **1616**) fully facing anterior. The retractor **1610** can be inserted in this position with all blades **1612**, **1614**, **1616** attached or with just the caudal and cephalad blades **1612**, **1614** attached initially. If added later, the anterior blade **1616** can be inserted through the working area and swept anteriorly under the great vessels to be attached to the retractor **1610**.

[0159] FIGS. **55A-55B** show an alternative version of retractor blade **1614** where docking pin sleeve **1624** is offset through the blade **1614**. The entry to the sleeve **1624** may include a chamfered or tapered surface **1626** around a portion or the entire periphery of the entry in order to enhance alignment of the pin **1650** with the sleeve. In this embodiment, the sleeve **1624** has a partially circular or semi-circular opening or channel that is in fluid communication with the interior of the blade **1614**. The sleeve **1624** includes at least a partial through hole extending through the inside of the blade **1614**. The sleeve **1624** is partially contained within the blade **1614**, thereby moving the bone pin **1650** closer to the working channel for improved visibility while still keeping the bone

pin **1650** out of the working window.

[0160] FIGS. **56A-56C** show another alternative embodiment for retractor blade **1614** including an insert **1628** which is at least partially received within sleeve **1624**, and FIGS. **57A-57B** show an embodiment of retractor blade **1614** including another version of insert **1628**. Unlike a traditional sleeve or shim which may slide down inside the blade occluding the surgeon's view and working area, one or more inserts **1628** may be embedded within the blade **1614**, thereby bringing the bone pin **1650** out of the working area and improving the viewing window.

[0161] The inserts **1628** may have any suitable size, shape, and configuration to mate the bone pin **1650** with the blade **1614**. In one embodiment, the insert **1628** has a semi-circular cross-section shape, for example, having an outer surface designed to mate with the channel or sleeve **1624** in the retractor blade **1614** and an inner surface designed to mate with and receive at least a portion of the bone pin **1650**. The insert **1628** may also include one or more openings, protrusions, or mating features, for example, on the inner or outer surfaces, configured to connect the insert **1628** to the blade and/or to the pin **1650**.

[0162] The insert **1628** can vary in length, to allow more engagement on the proximal and/or distal ends of the retractor blade **1614** as needed. In FIGS. **56A-56C**, the insert **1628** engages the proximal end of the retractor blade **1614** and is recessed within a portion of the sleeve **1624** of the blade **1614**. As shown, the insert **1628** extends less than halfway along the length of the sleeve **1624**. For example, the insert **1628** may extend about one third of the length of the blade **1614**. In FIGS. **57A-57B**, the insert **1628** is position along the entire length of the sleeve **1624**, and therefore engages the proximal and distal ends of the blade **1614**. The insert **1628** also engages a middle portion of the pin **1650**, having a reduced diameter as compared to enlarged portions on the proximal and distal ends of the pin **1650**. The enlarged proximal and distal ends of the pin **1650** may extend beyond the blade **1614**. The insert **1628** can be separate from the bone pin **1650** or may be permanently affixed to the bone pin **1650**. The insert **1628** may have substantially the same diameter as the bone pin **1650** to increase the cross sectional area of the retractor blade **1614**. A portion of the insert **1628** may extend past the proximal end of the blade **1614** such that the insert **1628** and/or the pin **1650** may be inserted and/or retrieved, for example, with a tool.

[0163] Various configurations of the retractor, blades, shims, inserts, and bone pins are contemplated and are not limited by the embodiments described with reference to the figures. For example, various sizes, shapes and types of retractors, blades, shims, inserts, and bone pins are contemplated, and various materials can be used to construct the various parts. For example, the retractor and its constituent components may be fabricated from any one or more, or combinations of, metals, polymers, carbon fiber or other composites, natural materials or any other materials having sufficient strength, durability and biocompatibility. Material selected can be radiolucent or radiopaque, as desired. The exemplary embodiments of the present disclosure provide various advantages, such as providing stability of the blades of the retractor, securing the retractor and preventing tissue creep.

[0164] All references cited herein are expressly incorporated by reference in their entirety. It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described herein above. In addition, unless mention was made above to the contrary, it should be noted that all of the accompanying drawings are not to scale. There are many different features to the present invention and it is contemplated that these features may be used together or separately. Thus, the invention should not be limited to any particular combination of features or to a particular application of the invention. Further, it should be understood that variations and modifications within the spirit and scope of the invention might occur to those skilled in the art to which the invention pertains. Accordingly, all expedient modifications readily attainable by one versed in the art from the disclosure set forth herein that are within the scope and spirit of the present invention are to be included as further embodiments of the present invention.

Claims

1. A retractor as substantially described herein.
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